Mittuniversitetet
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Försättsblad tentamen / Examination cover

2707183

Anonymitetskod / Anonymous code		
S - 0006 - SNK		
Kurskod / Course code: DT081A	Provkod/ Test code: T101	Tentamensdatum / Examination date: 2025-10-28
Kursnamn / Course name: Datateknik AV, Signal- och bildbehandling		
Provnamn / Test name: Tentamen		

Skriv din anonymitetskod på varje inlämnat papper
Write your anonymous code on each sheet submitted

Sätt ett kryss (x) för varje inlämnad uppgift
Use an x to indicate which questions has been submitted

Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes	Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes
1	X	5,5		16			
2	X	4		17			
3	X	4		18			
4	X	3,5		19			
5	X	3		20			
6	X	3		21			
7	X	5		22			
8	X	4		23			
9				24			
10				25			
11				26			
12				27			
13				28			
14				29			
15				30			
Poängsumma / Points		32	Betyg / Grade: B	Lärarsign./ Teachers sign			

Fylls i av tentamensvakt / To be filled in by the invigilator

Antal lösa blad/ No. of sheets submitted	Inlämnad tentamen / Submitted exam	Leg kontroll / Control identification	Sign. tentamensvakt / Sign. invigilator
10	0	✓	

2707183

Försättsbladet skall alltid lämnas in även om ingen uppgift behandlats
Examination cover should always be submitted even if no questions are answered

Utskriven 2025-10-24 kl. 12:45:00

DT081A – Signal and Image Processing, 2025-10-28

Solution examples

1. Definitions

a) **Sampling theorem:** A bandlimited signal can be perfectly reconstructed if sampled at a rate greater than twice its highest frequency.

b) **Aliasing:** Distortion caused by undersampling, where different frequency components become indistinguishable.

c) **Infinite Impulse Response (IIR):** A recursive system whose impulse response has infinite duration.

d) **Spectrum:** The frequency-domain representation of a signal.

e) **Digital signal:** A discrete-time signal with quantized amplitudes.

f) **Separability:** A property allowing a multidimensional operation to be implemented as successive one-dimensional operations.

2. System Classification and Difference Equation

a) FIR/IIR and causality

Feedback through a delay makes the system recursive, hence it is an **IIR** system. The presence of the delay ensures that only past output values are used, so the system is **causal**.

b) Difference equation

$$v[n] = x[n] + 0.3\,y[n-1], \qquad y[n] = 0.4\,v[n].$$

Combining the two equations yields

$$y[n] = 0.4\,x[n] + 0.12\,y[n-1].$$

3. System Analysis

Given

$$y(n) + y(n-1) = x(n) + 0.5\,y(n-2) - 0.3\,x(n-1),$$

under initial rest.

a) Type of response

For

$$x(n) = \begin{cases} 1, & n = 0, \\ 0, & \text{otherwise}, \end{cases}$$

the resulting output is the system's **impulse response** $h(n)$.

b) First three samples of the output

For

$$x[n] = \begin{cases} 0.6^n \cos(n), & n \geq 0, \\ 0, & \text{otherwise}, \end{cases}$$

the first three output samples are

$$\begin{aligned} y(0) &= 1, \\ y(1) &= 0.6\cos 1 - 1.3 \approx -0.96, \\ y(2) &= 0.36\cos 2 - 0.78\cos 1 + 1.8 \approx 1.23. \end{aligned}$$

c) Frequency response

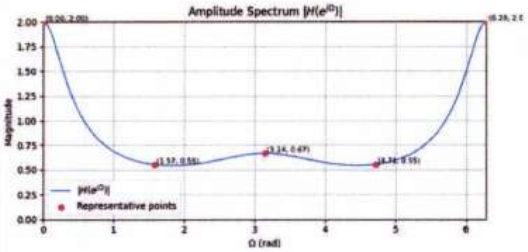
$$H(e^{j\Omega}) = \frac{1}{1 - e^{-j\Omega} - 0.5\,e^{-j2\Omega}}.$$

The magnitude response is

$$|H(e^{j\Omega})| = \frac{1}{\sqrt{(1 - \cos\Omega - 0.5\cos 2\Omega)^2 + (\sin\Omega + 0.5\sin 2\Omega)^2}}.$$

Representative values are

$$\begin{aligned} \Omega = 0: &\quad |H| \approx 2.0, \\ \Omega = \tfrac{\pi}{2}: &\quad |H| \approx 0.55, \\ \Omega = \pi: &\quad |H| \approx 0.67, \\ \Omega = \tfrac{3\pi}{2}: &\quad |H| \approx 0.55, \\ \Omega = 2\pi: &\quad |H| \approx 2.0. \end{aligned}$$



4. Sampling and Reconstruction

a) Lowest sampling frequency

Given

$$x(t) = 2\cos(60\pi t + 0.2) + 3\sin(30\pi t + 0.1),$$

the signal contains frequency components at 30 Hz and 15 Hz. By the Nyquist sampling theorem,

$$f_{\max} = 30 \text{ Hz}, \qquad f_s \geq 2 f_{\max} = 60 \text{ Hz}.$$

Thus, the minimum sampling frequency is $f_s = 60$ Hz.

b) Sampling at $f_s = 100$ Hz

The original frequency components are

$$15,\ 30 \text{ Hz}.$$

Aliased components due to spectral folding are

$$100 - 15 = 85 \text{ Hz}, \qquad 100 - 30 = 70 \text{ Hz}.$$

All components lie below the reconstruction low-pass filter cutoff, so the output spectrum contains

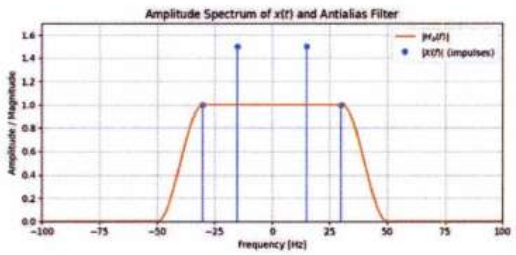
$$15,\ 30,\ 70,\ 85 \text{ Hz}.$$

c) Spectrum and antialias filter

The spectrum of $x(t)$ consists of impulses at

$$f = \pm 15,\ \pm 30 \text{ Hz}.$$

The antialiasing filter must pass frequencies up to 30 Hz, while its stopband should begin at $f_s/2 = 50$ Hz.



5. LTI System Response

a) Output signal

$$x(t) = 3\cos\!\left(10\pi t + \tfrac{\pi}{10}\right) + 1 + 4\sin(23\pi t).$$

From the magnitude and phase plots,

$$|H(0)| = 1, \quad \arg H(0) = \pi,$$

$$|H(10\pi)| \approx 1, \quad \arg H(10\pi) \approx 1.2,$$
$$|H(23\pi)| \approx 0.$$

Hence, the output signal is

$$y(t) = -1 + 3\cos\!\left(10\pi t + \tfrac{\pi}{10} + 1.2\right) + 0.5\sin(23\pi t).$$

b) Filter type

Low and high frequencies are passed with approximately unit gain, while a mid-frequency band is attenuated. This behavior corresponds to a **band-stop filter**.

6. Bayer Sampling and Demosaicing

a) Trade-off

Each pixel measures only one of $\{R, G, B\}$, so the remaining channels must be estimated from neighboring pixels. This interpolation acts as spatial low-pass filtering, which reduces spatial detail and may introduce color artifacts.

Higher color fidelity therefore comes at the cost of reduced spatial resolution, and vice versa.

b) Simple demosaicing

The Bayer pattern is periodic with a 2×2 RGGB structure, giving the ratio

$$G : R : B = 2 : 1 : 1.$$

At each pixel, the measured channel is kept and missing channels are interpolated from nearby samples:

- **Green pixel:** G known; R interpolated from nearest red neighbors in one direction; B interpolated from nearest blue neighbors in the orthogonal direction.
- **Red pixel:** R known; G averaged from 4-connected green neighbors; B averaged from 4 diagonal blue neighbors.
- **Blue pixel:** B known; G averaged from 4-connected green neighbors; R averaged from 4 diagonal red neighbors.

7. 2D Frequency Response Interpretation

a) Effect on spatial frequencies

The magnitude response $|H(u, v)|$ has a minimum near $(0, 0)$ and increases with radial frequency $\sqrt{u^2 + v^2}$. Low spatial frequencies are therefore attenuated, while high spatial frequencies are emphasized, corresponding to a high-pass or edge-enhancing filter.

b) Most likely output

- Image (i) is blurred, indicating low-pass filtering.
- Image (iv) preserves low frequencies and resembles the original, inconsistent with the shown frequency response.
- Images (ii) and (iii) are edge-dominant; image (iii) preserves a more complete set of edges.

The most likely output is therefore **image (iii)**.

8. Contrast Enhancement

Interpretation of plots

- **(e) Histogram** $h(r)$: Gray-level distribution of the original image, showing limited contrast.
- **(e) Cumulative histogram** $H(r)$: CDF of the original image.
- **(f) Cumulative histogram** $H'(s)$: CDF of the processed image, approximately linear after enhancement.
- **(d) Histogram** $h'(s)$: Gray-level distribution of the enhanced image, spread over a wider range.

The cumulative histogram $H(r)$ defines the intensity mapping

$$s = (L-1)\,H(r),$$

which, when applied to the original image, produces the contrast-enhanced result.

Lärarens anteckning / Teachers note:

1) a) SAMPLING THEOREM

~~It's the process of converting a continuous ti~~

IT'S THE PROCESS OF CONVERTING A CONTINUOS-TIME SINGAL (x(t)) INTO A DISCRETE-TIME SIGNAL, BY TAKING A SAMPLE OF x(t) EACH T[KNOWN AS PERIOD OF SAMPLING], $1/T = f_S$

THE ~~FNN~~ SAMPLING FUNCTION RESULTS AS:

$$x[m] = x(mT)$$

IT'S POSSIBLE TO SAMPLE CORRECTLY A SIGNAL IF $f_S \geq 2f_{max}$ THE SAMPLED SIGNAL HAVE TO CUTOFF $f_S/2$

b) ALIASING ~~OOO~~

ALIASING OCCURS WHEN THE SAMPLING FREQUENCY IS BELOW THE NSST RATE, $f_S < 2f_{MAX}$, (IT HAPPENS ANYWAY BUT IF $f_S \geq 2f_{MAX}$ IT DOESNT AFFECT THE SIGNAL BANDWIDTH, AND THE FREQUENCY CUTOFF GIVE THE SIGNAL CLEAN).

ALIASING WITH $f_S < 2f_{MAX}$ MEANS HIGHER FREQUENCIES ARE OUTSIDE OF THE BANDWIDTH OF THE SAMPLED SIGNAL, AFTER THE FREQUENCY CUTOFF, ($f_S/2$), AND THEY ARE FOLD BACK IN THE BANDWIDTH RESULTING IN NOISE AND DISTURBING THE SIGNAL, PLUS OBVIOUSLY NOT HAVING HIGHER FREQUENCY IN THE SIGNAL SAMPLED.

THE FORMULA ~~FOR~~ FIND ALIASING FREQUENCY IS

$$f_{alias} = |f - zf_S|, \quad z \in \mathbb{Z}$$

c) INFINITE IMPULSE RESPONSE (IIR)

FORMULA OF GENERAL FILTER: $y[m] = \sum_{K=1}^{N} a_K\, y(m-K) + \sum_{K=0}^{M} b_K\, x(m-K)$

~~IIR it's a filter with at least an ak coefficient ≠0~~,

IIR IT'S A FILTER WITH AT LEAST AN a_K COEFFICIENT $\neq 0$,

~~THIS~~ THIS MEANS CURRENT OUTPUT DOESNT ONLY DEPENDS ON THE CURRENT AND PREVIOUS INPUT LIKE FIR, BUT IIR FILTERS OUTPUT DEPENDS ALSO ON PREVIOUS OUTPUT, WHICH ARE CALLED FEEDBACKS.

IIR FILTER HAS AN INFINITE IMPULSE RESPONSE WHICH MEANS

$h(m) \neq 0 \quad -\infty < m < \infty$ (AFTER THE INPUT VALUE)

AND IF a_K COEFFICIENTS ARE MODELED WELL WE CAN OBTAIN

$$\lim_{m \to \infty} h(m) = 0$$

Lärarens anteckning / Teachers note:

1) d) SPECTRUM

SPECTRUMS ARE THE REPRESENTATION OF A SIGNAL IN THE FREQUENCY DOMAIN, IN FACT A SIGNAL $X(w)$ (OR $X(e^{jw})$) CAN BE SEEN AS

$$X(w) = |X(w)|\, e^{j \arg(X(w))}$$

$|H(w)|$: IS THE AMPLITUDE, OR MAGNITUDE (OFTEN REPRESENTED IN dB) WHICH GIVES INFORMATIONS ABOUT THE POWER OF THE SIGNAL IN EACH FREQUENCY COMPONENT

$\arg(X(w))$: IT'S THE ARGUMENT, OR PHASE WHICH GIVES INFOMATION ABOUT THE TIME DELAY OF EACH FREQUENCY COMPONENT.

$$|X(w)| = \sqrt{X(w) \cdot X^*(w)}\,, \quad \arg(X(w)) = \arctan\left(\frac{\mathrm{Im}(X(w))}{\mathrm{Re}(X(w))}\right) + (\pi \cdot \mathrm{flr}(X(w)))$$

THEY CAN BE VISUALIZED GRAPHICALLY IN 2D, with frequencies in the x axis, and amplitude or phase in the y axis.

e) DIGITAL SIGNAL

A SIGNAL IN ORDER TO BE STORED IN A COMPUTER HAS TO BE CONVERTED TO DIGITAL FROM ANALOGIC. THIS MEANS FIRST IT HAS TO BE SAMPLED, THEN IT HAS TO BE QUANTIZED DEPENDING ON HOW MANY LEVELS OF QUANTIZATION ARE CHOSEN, WITH TRADEOFF OF ROUNDOFF. THIS LEADS TO A DIGITAL REPRESENTATION OF THE SIGNAL STORED IN COMPUTERS AS A SEQUENCE OF BITS.

f) SEPARABILITY

THIS PROPERTY ALLOWS TO SEPARATE N^x, $x \geq 2$ INTO EACH RESPECTIVE BASE ($N \times N \times \ldots$)

AN EXAMPLE FOR SEPARABILITY IS THE DCT OPERATION:

$$F(u,v) = \frac{2\,C(u)C(v)}{\sqrt{MN}} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} \cos\left(\frac{2i+1}{2M} \cdot v\pi\right) \cdot \cos\left(\frac{2j+1}{2N} \cdot v\pi\right) \cdot f(i,j)$$

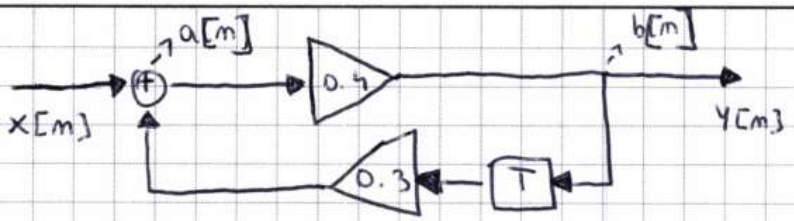
repeated in his 2 bases (x,y), which allows more efficient computation:

$$G(i,v) = \sqrt{\frac{2}{N}}\, C(v) \sum_{j=0}^{N-1} \cos\left(\frac{2j+1}{2N} v\pi\right) \cdot f(i,j)$$

$$F(u,v) = \sqrt{\frac{2}{M}}\, C(u) \sum_{i=0}^{M-1} \cos\left(\frac{2i+1}{2M} u\pi\right) \cdot G(i,v)$$

Lärarens anteckning / Teachers note:

2)



a) IT'S IIR BECAUSE CURRENT OUTPUT DEPENDS ON CURRENT INPUT AND ON PREVIOUS OUTPUT, SO $a_K \neq 0$ FOR SOME K.
IT'S CAUSAL BECAUSE OUTPUT DOESN'T DEPEND ON FUTURE EVENTS, SO CAUSE ARRIVES BEFORE THE EFFECT, AND THE IMPULSE RESPONSE $h(m)$ CAN BE DEFINED AS:

$$h(m) = \begin{cases} h^+(m) & m \geq 0 \\ 0 & m < 0 \end{cases}$$

THE SYSTEM IS ALSO LTI BECAUSE USES ONLY LINEAR OPERATIONS SUCH AS SUM AND MULTIPLICATION BY CONSTANTS, AND T FUNCTION IS A CONSTANT DELAY, SO SYSTEM DOESN'T DEPEND ON TIME.

SO THIS IS A TIME DISCRETE A LTI CAUSAL IIR SYSTEM.

b)

$b[m] = y[m]$, $b[m] = 0.4\, a[m]$, $b[m-1] = y[m-1]$

$$a[m] = x[m] + 0.3\, b[m-1] = x[m] + 0.3\, y[m-1]$$

So $y[m] = 0.4\, a[m]$, $\frac{y[m]}{0.4} = a[m]$

$$\frac{y[m]}{0.4} = x[m] + 0.3\, y[m-1]$$

$$y[m] = 0.4\, x[m] + 0.4 \cdot 0.3\, y[m-1]$$

$$y[m] = 0.4\, x[m] + 0.12\, y[m-1]$$

Lärarens anteckning / Teachers note:

3) $y(m) + y(m-1) = x(m) + 0.5\,y(m-2) - 0.3\,x(m-1)$

~~$y(m) = x(m) - 0.3x(m-1) + 0.5$~~

$y(m) = x(m) - 0.3x(m-1) - y(m-1) + 0.5\,y(m-2)$

a) Since $x(m) = \delta[m]$ IS AN IMPULSE INPUT

THE SYSTEM PRODUCES AN UNIT PULSE (IMPULSE) RESPONSE

$h[m]$ WHICH DESCRIBES FULLY THE BEHAVIOUR OF THE SYSTEM.

b) $x[m] = \begin{cases} 0.6^m \cdot \cos[m], & m \geq 0 \\ 0, & \text{otherwise} \end{cases}$

OUTPUT $y(0), y(1), y(2)$?, SINCE THE SYSTEM IS CAUSAL $y(m)=0$, $m<0$

$y(0) = 0.6^0 \cos(0) - 0.3 \cdot 0 - 0 + 0.5 \cdot 0 = 0.6\cos(0) = 0.6$

$y(1) = 0.6^1 \cos(1) - 0.3 \cdot 0.6 \cos(0) - 0.6 + 0.5 \cdot 0 = 0.6\cos(1) - 0.18 - 0.6$

$y(1) = 0.6\cos(1) - 0.78 \approx 0.32 - 0.78 = -0.46$

"if cos is in radiant then

$y(2) = 0.6^2 \cos(2) - 0.3 \cdot 0.6\cos(1) - 0.6\cos(1) - 0.78 + 0.3 =$

$= 0.36\cos(2) - 0.78\cos(1) - 0.48 \approx -1.05$

if cos is in radiant then

c) $H(e^{j\omega}) = \dfrac{e^{2j\omega}}{e^{2j\omega} - e^{j\omega} - 0.5}$

$e^{j0} = 1$, $e^{j\pi/2} = j$, $e^{\pi} = -1$, $e^{3\pi/2} = -j$, $e^{2\pi} = 1$

$e^{j\pi/4} = \frac{1}{4}\pi = \sqrt{2}/2 + j\frac{\sqrt{2}}{2}$

$e^{\frac{3}{4}\pi} = -\sqrt{2}/2 + j\frac{\sqrt{2}}{2}$

$H(0) = \dfrac{e^0}{e^0 - e^0 - 0.5} = \dfrac{1}{1 - 1 - 0.5} = -\dfrac{1}{0.5}$, $|H(0)| = \dfrac{1}{0.5} = 2$

$H(\pi/2) = \dfrac{e^{j\pi}}{e^{j\pi} - e^{j\pi/2} - 0.5} = \dfrac{-1}{-1 - j - 0.5}$, $|H(\pi/2)| = \dfrac{\sqrt{(-1)^2}}{\sqrt{(-1.5)^2 + (-1)^2}} =$

$\dfrac{1}{\sqrt{2.25 + 1}} \approx 0.55$

$H(\pi/4) = \dfrac{e^{j\pi/2}}{e^{j\pi/2} - e^{j\pi/4} - 0.5} = \dfrac{j}{j - \frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2} - 0.5}$

$|H(\pi/4)| = \sqrt{\quad + 0.5^2}$

$= 0.805067 \approx 0.8$

Lärarens anteckning / Teachers note:

3) c)

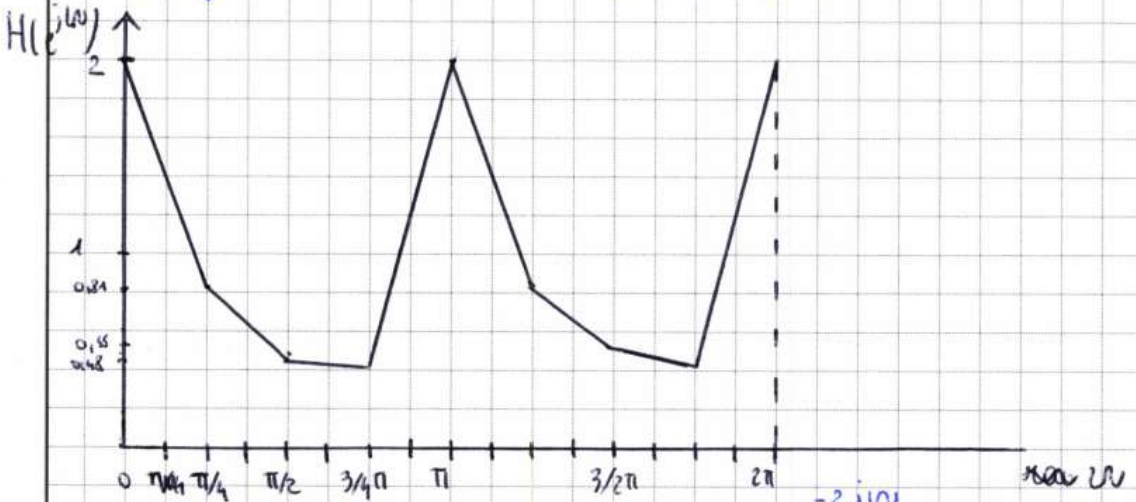
$H(\pi) = \frac{e^{j2\pi}}{e^{j2\pi} - e^{j\pi} - 0,5} = \frac{1}{1 - 1 - 0,5} = \frac{1}{0,5}$, $|H(\pi)| = \frac{1}{0,5} = 2$

$H(\frac{3}{2}\pi) = \frac{e^{j3\pi}}{e^{j3\pi} - e^{\frac{3}{2}\pi} - 0,5} = \frac{-1}{e^{j\frac{3}{2}\pi} + 1 + j - 0,5}$, $\frac{-1}{-1,5+j}$,

$|H(\frac{3}{2}\pi)| = \frac{1}{\sqrt{3,25}} \approx 0,55$

$H(2\pi) = \frac{e^{j4\pi}}{e^{j4\pi} - e^{j2\pi} - 0,5} = \frac{1}{1 - 1 - 0,5}$, $-\frac{1}{0,5}$, $H(2\pi) = \frac{1}{0,5} = 2$

the signal seem to have a periodicity of π



$|H(e^{j\omega})| = \frac{e^{2j\omega}}{e^{2j\omega} - e^{j\omega} - 0.5}$ $\frac{e^{-2j\omega}}{e^{-2j\omega} - e^{-j\omega} - 0,5}$

$\frac{1}{1 + 1 + 0.25 - e}$

Skriv ej på baksidan! Do not write on the other side!

Lärarens anteckning / Teachers note:

4) $x(t) = 2\cos(60\pi t + 0.2) + 3\sin(30\pi t + 0.1)$

Knowing that a periodic signal is written as:

$x(t) = a_1\cos(2\pi \cdot f_1 t + \phi_1) + a_2\sin(2\pi \cdot f_2 \cdot t + \phi_2) \ldots$

rewriting $x(t) = 2\cos(2\pi \cdot 30 \cdot t + 0.2) + 3\sin(2\pi \cdot 15 \cdot t + 0.1)$

a) $f_s \geq 2 f_{max}$, f_{max} FOR $x(t)$ is 30 Hz

$\Rightarrow f_s = 2 \cdot 30$ Hz $\Rightarrow f_s = 60$ Hz

b) ALIASING IN THIS CASE OCCURS BECAUSE THE SAMPLED SIGNAL SHOUL BE CUTOFF AT 30 Hz BUT IT HAS CUTOFF AT $f_s = 100$ Hz. SO THERE ARE SOME ALIASING FREQUENCY BETWEEN f_{max} AND $f_s/2$ ($f_s/2 = 50$ Hz) AND F_c

folding $|f - z f_s|$ case $f = 15$ wire $f = 30$

$z = 1 \quad |15 - 1 \cdot 100| = 85$

$z = 1 \quad |30 - 1 \cdot 100| = 70$

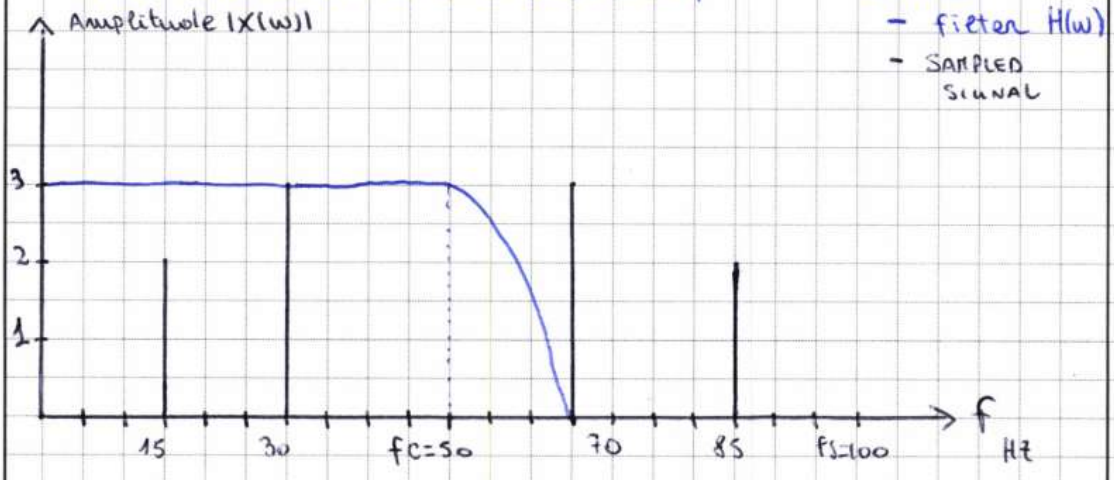
outside of the signal: $z = -1 \quad |30 + 1 \cdot 100| = 130$

outside of the signal: $z = -2 \quad |30 - 2 \cdot 100| = 170$

THE OUTPUT SIGNAL CONTAINS THE FOLLOWING FREQUENCIES

frequencies of $\tilde{x}_{LP}(t) =$ [15, 30, 70, 85] Hz

c) ASSUMING it's still sampled at $f_s = 100$ Hz, an antialising filter (for this specific case), SHOULD HAVE A FREQUENCY CUTOFF AT 50 Hz, (SHOULD BE BETTER $F_c = 30$ Hz TO AVOID EACH POSSIBLE ALIASING). FOR THE REAL FILTER I CHOOSE A BUTTERWOOTH BECAUSE HAS NO RIPPLE IN THE FLAT BAND (PASS BAND)



IT'S FUNDAMENTAL THAT THE FILTER GOES TO 0 BEFORE THAT THE ALIASING COME (70 Hz) SO I SUGGEST TO REDUCE THE F_c TO AVOID THIS RISK.

Lärarens anteckning / Teachers note:

5) $x(t) = 3\cos(10\pi t + \frac{\pi}{10}) + 1.4\sin(23\pi t)$

$Y(\omega) = |H(\omega)| e^{j\arg(H(\omega))} \cdot X(\omega)$

a)

$\omega = \text{rad}/s \rightarrow 2\pi \cdot f$

$-\cos \rightarrow \omega \rightarrow 10\pi \approx 31.4\ \omega$

$|H(\omega)|_{\omega = 31.4} = 1$

$\arg(H(\omega)) = \begin{cases} \pi - \frac{\omega \cdot \pi}{50}, & 0 \le \omega \le 50 \\ 0, & 50 \le \omega \le 75 \\ \pi - \frac{\omega - 75}{50}\pi, & 75 \le \omega \le 125 \end{cases}$

$\arg(H(\omega))|_{\omega = 31.4} = 0.372\pi$

$\downarrow$

$3.1\cos(10\pi t + \frac{\pi}{10} + 0.372\pi)$

$-\sin \rightarrow \omega \rightarrow 23 \cdot 3.14 \approx 72.22\ \omega$

$|H(\omega)|_{\omega = 72.22} \approx \frac{1}{8}(0.125)$, $\arg(H(\omega))|_{\omega = 72.22} = 0$

$\downarrow$ $+1.4 \cdot \frac{1}{8} \cdot \sin(23\pi t + 0)$

$-1 \Rightarrow 1\cos(2\pi \cdot 0 \cdot t) \rightarrow \omega = 0$

$|H(\omega)|_{\omega=0} = 1$, $|\arg(H(\omega))|_{\omega=0} \pi \rightarrow \cos(\pi) = -1$

So $y(t) = h(t) * x(t) =$

$$\boxed{y(t) = 3\cos(10\pi t + \frac{\pi}{10} + 0.372\pi) + \frac{1}{2}\sin(23\pi t) - 1}$$

b) THE SYSTEM IS A FILTER SIMILAR TO A STOP-BAND FILTER, BUT INSTEAD OF CANCELING COMPLETELY THE FREQUENCIES BETWEEN ω_1 AND ω_2 (≈ 50 AND ≈ 118.5 RESPECTELY) IT ATTUANES THEM REDUCING THE AMPLITUDE UNTIL ≈ 0.125. INFACT SIN WHICH IS IN BETWEEN ω_1 AND ω_2 HAS HITS AMPLITUDE DIVIDED BY A FACTOR OF 8. PLUS THERE IS A PHASE SHIFT ~~OF $+\pi$ BETWEEN~~ THAT LINEARLY DECREASE FROM π TO 0 IN $[0, 50]$ AND ANOTHER PHASE SHIFT THAT LINEARLY DECREASE FROM 0 TO $-\pi$ IN $[75, 125]$

THE AMPLITUDE OF FREQUENCY $< \omega_1$ AND $> \omega_2$ (IN THE SIMILAR PASS BAND) REMAIN THE SAME (FACTOR OF 1).

SO THIS FILTER SPECIFICALLY ATTENUATE (NOT CANCEL) A SPECIFIC PORTION OF THE FREQUENCY, BETWEEN ω_1 AND ω_2.

Lärarens anteckning / Teachers note:

6)

a) ~~BECO~~ BECAUSE EACH PIXEL ONLY CAPTURES ONE OF THE THREE COLOR COMPONENT (RED, GREEN OR BLUE) WHICH MEANS THAT THE PIXEL DOESN'T KNOW THE EXACTLY VALUE OF THE OTHER TWO COMPONENTS IN THAT SPECIFIC LOCATION, BUT IT INTERPOLATES THEM FROM ~~THE~~ NEIGHBORING PIXELS. THIS ALLOWS CAMERA TO HAVE JUST ONE LAYER FOR THE THREE COMPONENT DESPITE HAVING A LOWER COLOR FIDELITY. IT'S IMPORTANT SO TO HAVE MANY PIXELS IN THE CAMERA SENSOR IN ORDER TO OBTAIN A GOOD IMAGE, OTHERWISE THERE IS THE RISK OF NEIGHBORING PIXELS REFERING ~~TO~~ ALWAYS TO DIFFERENT COLORS.

b) Since the matrix contained

G R G R G R ... so more greens, because our
B G B G B G ... eyes are more sensible to
G R G R G R ... green
. . . . . .

THE GREEN PIXEL ~~VALUE~~ IS

G R G
B G B , FOR THE RED COMPONENT
G R G TAKES THE AVERAGE OF THE TWO RED

THE RED PIXEL ~~VALUE~~

B G B
G R G
B G B

~~THE~~ BLUE PIXEL

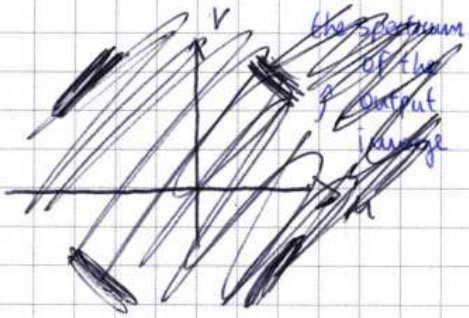
R G R
G B G
R G R

~~two interpolated the other pixel value, the average~~

TO INTERPOLATED THE OTHER PIXELS VALUE EACH PIXEL CALCULATES THE AVERAGE VALUE OF ITS NEIGHBORING PIXELS WITH THE COMPONENT IS LOOKING FOR (IF IT IS LOOKING FOR RED JUST THE RED, IF IT IS LOOKING FOR BLUE JUST THE BLUE, SAME FOR GREEN).

Lärarens anteckning / Teachers note:

7)



a) ~~low freq~~

LOW FREQUENCY NEAR TO THE CENTER (0,0) ARE CANCELED THIS HAS AS CONSEQUENCE THAT THE PART OF THE IMAGE WHERE CONTRAST IN'S NOT PRESENT ARE CANCELED (REGION WITH SAME COLOR) ~~SO~~.

~~THE FREQUENCY AND COMPONENTS V ARE GETTING INCREASED HIGHER AND HIGHER DEPENDING ON THE FLOW~~

~~IN~~ AS THE FREQUENCY GO FAR FROM THE CENTER THE AMPLITUD IS GETTING MULTIPLIATE TO A FACTOR HIGHER AND HIGHER.

IN $F(0,-0.4)$, $F(0,0.4)$, $F(0.4,0)$, $F(-0.4,0)$ it stops $\approx 4/5$

while in the FURTHEST REGION $F(0.4,0.4)$, $F(0.4,-0.4)$, $F(-0.4,0.4)$, $F(-0.4,-0.4)$ IT ARRIVES AT 7

THIS SYSTEM IS SIMILAR TO AN HIGH PASS FILTER BECAUSE CANCEL LOW FREQUENCIES, BUT INSTEAD OF PRESERVING OTHERS THE FILTER INCREAS THE REMAING FREQUENCIES BY A FACTOR INCREASING DEPENDING HOW MUCH THE FREQUENCIES ARE TO THE CENTER. THIS SHOULD ALLOWS TO SHOW EDGE, BUT WITH SOME DIFFERENCES FROM A NORMAL H-P FILTER

b) i) NO BECAUSE IT'S A BLUR, WHICH TAKE THE AVERAGE OF NEIGHBOURHOOD PIXELS (USUALLY), LOW FREQUENCY ARE PRESERVED DESPITING HIGHEST THAT ARE ATTENUATED/REMOVED. H-P= HIGH-PASS

ii) THIS COULD BE, BECAUSE ONLY SHOWS EDGE (PRESERVED BY H-P FILTER) BUT IT'S SEEMS LIKE AN HORIZONTAL EDGE DETECTOR, WHILE THE SYSTEM SHOWN IN FIGURES ($|H(u,v)|$) PRESERVES ALSO THE VERTICAL ~~ONES~~ EDGES. SO NO

iii) YES, BECAUSE THIS IMAGE SHOWS ONLY THE EDGE OF THE IMAGE AND EACH TYPE OF EDGE (VERTICAL, HORIZONTAL) IS TREATHED IN THE SAME WAY. THAT'S WHAT A SIMILAR HIGH-PASS FILTER LOOKS LIKE

iv) NO, BECAUSE THIS IMAGE JUST INCREASE THE BRIGHTNESS OF THE IMAGE BY A CONSTANT, PLUS COLORS ARE PRESERVED.

SO THE IMAGE iii) CORRISPOND TO THE FILTER

Lärarens anteckning / Teachers note:

8) THE IMAGE (b) IS THE IMAGE (a) AFTER AN HISTOGRAM EQUALIZATION

c) AND d) IMAGES SHOW THE HISTOGRAM (PDF), THAT SHOWS HOW MANY PIXELS HAVE THE SAME INTENSITY VALUES *

d) IS THE c) HISTOGRAM AFTER THE HISTOGRAM EQUALIZATION *

e) IMAGE IT'S THE HISTOGRAM CDF OF THE (a) IMAGE

HISTOGRAM CDF: $H(x \leq a) \sum_{j=0}^{a} h(j)$ ↓ HISTOGRAM

PROBABILITY THAT A PIXEL HAS AN INTENSITY ↓ VALUE BELOW a

f) IT'S THE HISTOGRAM CDF AFTER THE EQUALIZATION, THE CDF IT'S UNIFORMLY DISTRIBUTED AMONG EACH INTENSITY VALUE. THIS MEANS EACH INTENSITY VALUE (OR SCALE OF GRAY) IS USED AT THE SAME FREQUENCY AMONG THE PIXELS OF THE IMAGE (b).
UNIFORM CDF ALLOWS TO ENHANCE CONTRAST

IMAGE I' (b) IS OBTAINED FROM IMAGE I (a) USING H' ~ UNIFORM CDF → CDF UNIFORMLY DISTRIBUTED!

$$I' = H' \cdot I$$

* c) IS THE HISTOGRAM OF THE IMAGE (a) AND d) IS THE HISTOGRAM OF THE IMAGE (b)